

Advanced Spectroscopic Analysis of Organic Compounds

Science

May, 2026



Advanced Spectroscopic Analysis of Organic Compounds (A32061)

Short Title:	Adv Spectro Analysis Org Cmpds	
Faculty	Science and Computing	
Department	Science	
Cluster	Analytical Science	
Author	KMURPHY	
Credits	5	Level: Intermediate

Description of Module / Aims

This twelve week intensive module will cover the use of UV-Vis, nuclear magnetic resonance (NMR) and infra-red (IR) spectroscopy and mass spectrometry (MS) in the structural elucidation of organic molecules.

Programmes

CHEM-0038	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	3 / 5 / M
CHEM-0038	BSc in Pharmaceutical Science (WD_SPHSC_D)	3 / 5 / M

Indicative Content

- Infrared spectroscopy: prediction of major absorbance bands from chemical structural elements in molecules
- Mass spectrometry: basic principles, instrumentation, isotopes and fragmentation patterns
- NMR spectroscopy: basic principles, instrumentation, detailed interpretation of proton and carbon-13 NMR spectra
- Use of UV-Vis, NMR, IR and mass spectra in structural elucidation of organic molecules
- Application of techniques in the characterisation of synthesised products and unknowns in laboratory practicals

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Determine the structure of an organic compound from an analysis of combined spectral data.
2. Relate the theory of the techniques of NMR and MS to the characterisation of organic compounds.
3. Analyse the different spectroscopic techniques used in organic chemistry.
4. Use practical techniques to determine the structure of a synthesised product and an unknown structure.

Learning and Teaching Methods

- Lectures.
- Laboratory-based practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Tutorial/Problem Sheets</i>	20%	1,2,3
<i>In-Class Assessment</i>	30%	1,2
<i>Lab Report</i>	50%	4

Assessment Criteria

- <40%:** Little or no demonstration of an understanding of the fundamentals of the advanced spectroscopic analysis of organic compounds. Lack of competence with regard to associated laboratory skills, practices and instrumentation.
- 40%-49%:** Will be able to show a limited understanding of the fundamentals of the advanced spectroscopic analysis of organic compounds, but may have some difficulty with applying this theory. Will display some ability with regard to associated laboratory skills, practices and instrumentation.
- 50%-59%:** As well as the above, the student will be to show a reasonable understanding of the complexities of the advanced spectroscopic analysis of organic compounds, and predict some trends in analytical results. Will display reasonable competence with regard to associated laboratory skills, practices and instrumentation.
- 60%-69%:** In addition, the student will demonstrate a more detailed knowledge of the advanced spectroscopic analysis of organic compounds and show ability to analyse problems in this area. Will display very good competence with regard to associated laboratory skills, practices and instrumentation.
- 70%-100%:** All of the above to an excellent level. In addition, the learner will be able to demonstrate comprehensive knowledge and understanding of the advanced spectroscopic analysis of organic compounds, predict general trends in behaviour, including exceptional behaviour and critically evaluate and relate topics. Will display advanced competence and independence with regard to associated laboratory skills, practices and instrumentation.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the theory and practical components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Lab	36	
Independent Learning	75	

Essential Material

- "Mastering Chemistry." www.pearsonmylabandmastering.com/masteringchemistry
- Bruice, P. _Organic Chemistry_. 8th ed. New Jersey: Pearson, 2010.

- Williams , D.H. and I. Fleming. _Spectroscopic Methods in Organic Chemistry_. 5th ed. London: Mc Graw Hill, 1995.

Supplementary Material

- "Virtual Textbook of Organic Chemistry." <http://www.cem.msu.edu/~reusch/vtxtindex.htm>
- Richards, S.A. _Essential Practical NMR for Organic Chemistry_. Palo Alto, Calif. : John Wiley & Sons, 2011.

Requested Resources

- Room Type: Chemistry Lab
- Lecture Room: Loose Seated